

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3.	(a)			In a <u>water</u> bath or in a beaker of <u>water</u> must imply heading (1) Using ice cubes / taking water from fridge / in an ice bath or equivalent (1)	1		1	2		2
	(b)	(i)		Straight line of best fit drawn (from 18 – 100 °C) with points either side or reference to very little scatter in the readings (1) Reference to or comparison with $y = mx + c$ (1) Positive intercept or positive gradient (1)		1	1 1	3		3
		(ii)	I	5.5 Ω , 5.6 Ω or 5.7 Ω unit mark [Accept attempts at 3 sf]		1		1	1	1
			II	Either Readings from graph line inserted in $\frac{\Delta R}{\Delta \theta}$. Tolerate slips (1) Gradient = 0.023 [± 0.002] [$\Omega ^\circ\text{C}^{-1}$] or by implication (1) $\alpha = 3.9 [\pm 0.4] \times 10^{-3} \text{ }^\circ\text{C}^{-1}$ unit mark ecf on R_0 and gradient (1) Or Value of R_0 and (R, θ) values from another point on graph line inserted in equation. Tolerate slips (1) Correct algebra (1) $\alpha = 3.9 [\pm 0.4] \times 10^{-3} \text{ }^\circ\text{C}^{-1}$ unit mark ecf on R_0			1 1 1	3	3	3
				Question 3 total	1	2	6	9	4	9